

Bayesian inference on the Planck 2018 binned temperature (TT) power spectrum

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Abstract

We perform a simplified but real Bayesian reanalysis of the Planck 2018 binned temperature (TT) angular power spectrum. The publicly released bandpowers are compared with a theoretical C_ℓ^{TT} computed with CAMB for the six base Λ CDM parameters, using a per-bin Gaussian likelihood (diagonal covariance) and a Gaussian prior on the optical depth τ . The posterior is sampled with the affine-invariant ensemble sampler **emcee**. We recover the Λ CDM parameters within $\sim 1\text{--}2\sigma$ of the official Planck values, obtain a good fit ($\chi^2/\text{dof} = 0.94$), and explicitly show that the A_s constraint is inherited from the τ prior through the $A_s e^{-2\tau}$ degeneracy. Convergence, prior sensitivity, residuals, and multipole-range consistency are assessed. The work reproduces the spirit of the Planck parameter inference while remaining light enough to run on a laptop.

1 Introduction and background

The cosmic microwave background (CMB) is the relic radiation released when the Universe became transparent, about 380,000 years after the Big Bang. Tiny temperature anisotropies imprinted on it encode the physics of the primordial plasma: acoustic oscillations driven by the competition between gravity and radiation pressure produce a characteristic series of peaks in the angular power spectrum C_ℓ^{TT} , whose positions and heights depend sensitively on the cosmological parameters. Measuring this spectrum and inferring those parameters is one of the central tasks of modern cosmology, and the Planck satellite has measured the temperature spectrum to high precision across a wide range of multipoles ℓ [1].

The task addressed here is a parameter-estimation problem: given the measured (binned) bandpowers $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ and their uncertainties, infer the parameters of the standard six-parameter Λ CDM model. This is naturally a *Bayesian* problem. First, the measurement carries genuine uncertainty, and we want not a single best-fit point but the full posterior distribution of the parameters, including credible intervals and the correlations (degeneracies) between parameters. Second, external information is most consistently incorporated as a prior: the optical depth τ is poorly constrained by temperature data alone, and a Gaussian prior derived from the Planck low- ℓ polarisation measurement is the principled way to include that knowledge. Third, the Bayesian framework extends cleanly to model comparison and to propagating measurement error into derived quantities. A plain χ^2 minimisation would return a point estimate but none of this structure; Bayesian inference with Markov-chain Monte Carlo (MCMC) sampling is therefore the standard tool for cosmological parameter estimation [4].

We deliberately work with a simplified version of the full Planck analysis. We use the publicly released CMB-only *binned* TT bandpowers rather than the full multipole-by-multipole data; we adopt a Gaussian likelihood that keeps only the per-bin variance, ignoring the bin-to-bin and pixel-level correlations of the official Planck likelihood; and we model only the temperature spectrum, without polarisation or the detailed treatment of residual foregrounds. These choices make

the analysis light enough to run on a laptop while remaining a faithful, end-to-end reproduction of the inference: real data, a real Boltzmann theory code (CAMB [3]), a real likelihood, and a real MCMC sampler (`emcee` [2]). The consequences of the simplifications are discussed where they matter, particularly in Section 4 and in the interpretation of the results in Section 5.

2 Data

We use the publicly released Planck 2018 binned temperature power spectrum, distributed as the plain-text file `COM_PowerSpect_CMB-TT-binned_R3.01.txt` in the Planck Public Data Release 3 ancillary data [1]. The file contains the CMB-only binned bandpowers, marginalised over astrophysical foregrounds and released for plotting and inspection purposes. It has five whitespace separated columns:

Column	Meaning
ℓ	effective (weighted) multipole of the bin
D_ℓ	bandpower $D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell$ [μK^2]
$-dD_\ell$	lower 1σ uncertainty on D_ℓ
$+dD_\ell$	upper 1σ uncertainty on D_ℓ
BestFit	Planck best-fit ΛCDM bandpower

The spectrum spans roughly $\ell \simeq 30$ to $\ell \simeq 2500$ in a few tens of bins. Throughout we work directly with D_ℓ in μK^2 and treat each bin’s uncertainty as independent (diagonal covariance); the consequences of this simplification are discussed in Section 4. Figure 1 shows the loaded data.

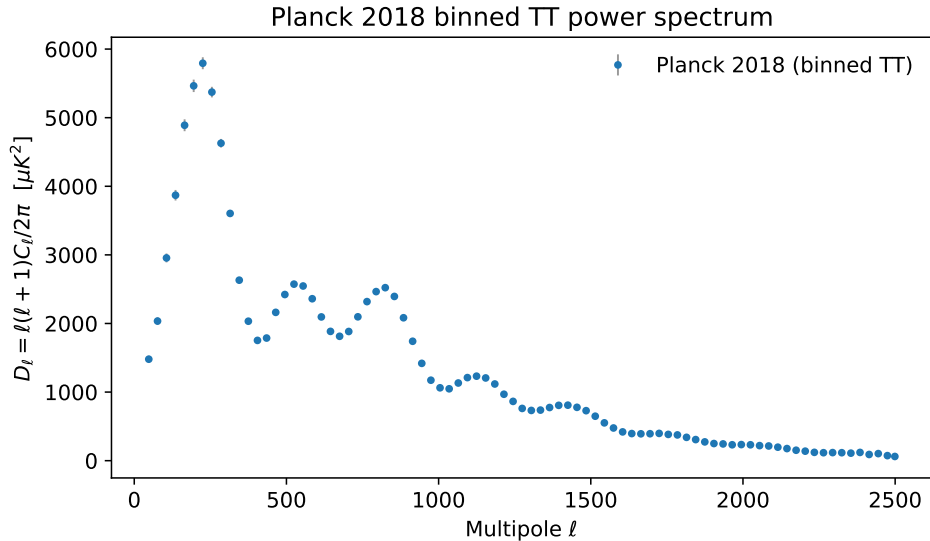


Figure 1: Planck 2018 binned TT bandpowers D_ℓ with 1σ error bars.

3 Theoretical model

The theoretical TT power spectrum is computed with the Boltzmann code CAMB [3] through its Python interface (the `camb` package, installed via `pip`; no manual compilation is required). The model is parametrised by the six base ΛCDM parameters

$$\theta = (\Omega_b h^2, \Omega_c h^2, H_0, \tau, \ln(10^{10} A_s), n_s), \quad (1)$$

where $\Omega_b h^2$ and $\Omega_c h^2$ are the physical baryon and cold dark matter densities, H_0 the Hubble constant, τ the reionisation optical depth, A_s the amplitude and n_s the tilt of the primordial scalar power spectrum. We sample $\ln(10^{10} A_s)$ rather than A_s because it varies over a more comparable numerical range.

For a given θ , CAMB returns the lensed spectrum $D_\ell^{TT} = \ell(\ell+1)C_\ell^{TT}/2\pi$ in μK^2 at integer multipoles. To compare with the binned data we evaluate the theory at the *effective* multipoles ℓ_b of each bin by linear interpolation,

$$D_b^{\text{th}}(\theta) = D_\ell^{TT}(\theta) \Big|_{\ell=\ell_b}, \quad (2)$$

which approximates the Planck bin window functions (not provided in the public binned file) and is adequate because D_ℓ^{TT} varies smoothly across each bin.

Two deliberate simplifications are adopted for computational speed, both negligible at the precision of this simplified analysis: neutrinos are treated as massless ($\sum m_\nu = 0$ instead of the default 0.06 eV), and the lensing potential is computed at low accuracy. Figure 2 overlays the CAMB model evaluated at the Planck 2018 fiducial parameters on the data as a sanity check.

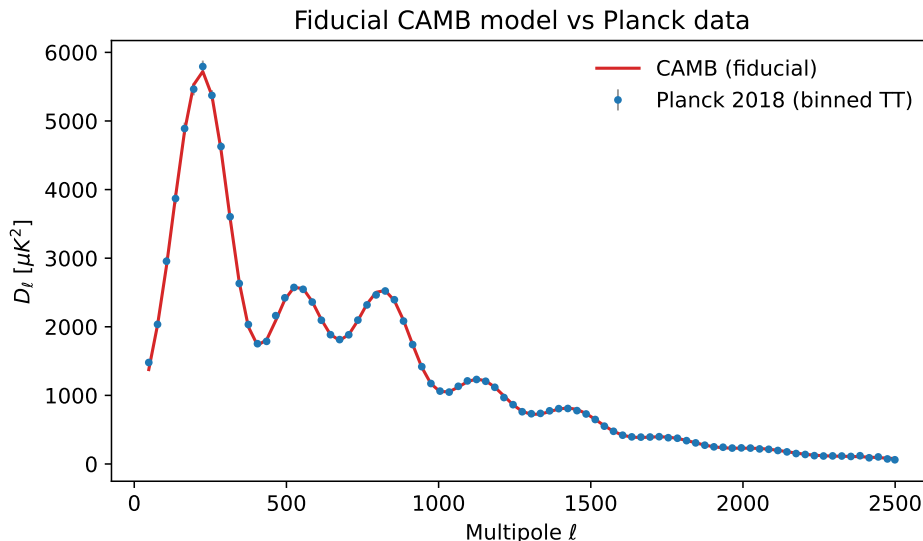


Figure 2: CAMB model at the Planck 2018 fiducial parameters overlaid on the binned TT data.

4 Methodology

4.1 Likelihood

We adopt a deliberately simplified Gaussian likelihood that retains only the per-bin variance and ignores the bin-to-bin (and pixel-level) correlations of the full Planck likelihood:

$$\ln \mathcal{L}(\theta) = -\frac{1}{2} \sum_b \frac{(D_b^{\text{data}} - D_b^{\text{th}}(\theta))^2}{\sigma_b^2}, \quad (3)$$

where D_b^{data} and σ_b are the bandpower and its symmetrised 1σ uncertainty from Section 2, and $D_b^{\text{th}}(\theta)$ is the binned theory of Section 3. The additive normalisation constant is dropped as it does not affect the sampling. Treating the covariance as diagonal slightly *over*-states the information content relative to the official likelihood; we therefore interpret the resulting credible intervals as indicative rather than definitive, and revisit this in Section 5.

4.2 Priors

We use uniform priors on five parameters over physically reasonable ranges (Table 1) and a Gaussian prior on the optical depth,

$$\tau \sim \mathcal{N}(0.0544, 0.0073^2), \quad (4)$$

taken from the Planck low- ℓ EE measurement. As discussed in Section 1, TT-only data constrain A_s and τ only through the combination $A_s e^{-2\tau}$, so this prior is what breaks the degeneracy; we expect the τ posterior to essentially reproduce the prior.

Parameter	Prior	Range / (μ, σ)
$\Omega_b h^2$	uniform	[0.018, 0.026]
$\Omega_c h^2$	uniform	[0.10, 0.14]
H_0	uniform	[55, 80]
τ	Gaussian	(0.0544, 0.0073)
$\ln(10^{10} A_s)$	uniform	[2.7, 3.3]
n_s	uniform	[0.92, 1.00]

Table 1: Prior specification for the six sampled parameters.

The log-posterior is $\ln P(\theta | \text{data}) = \ln \mathcal{L} + \ln \pi(\theta)$ up to a constant. Evaluations of the theory are wrapped so that parameter values for which the Boltzmann solver fails return $-\infty$, keeping the chains numerically robust.

4.3 Sampler

The posterior is explored with the affine-invariant ensemble sampler `emcee` [2]. We use $N_w = 24$ walkers (four per dimension), initialised in a tight Gaussian ball around the Planck 2018 fiducial point, each within the prior box. The chain is run for a burn-in phase of 200 steps, which is then discarded, followed by a production phase of 1500 steps that is retained for analysis.

Because each likelihood call is dominated by a CAMB evaluation, the walker updates are parallelised with a process pool, giving a near-linear speed-up in the number of cores. The model and likelihood are packaged in a small module (`planck_model.py`); the pool uses the `spawn` start method, so each worker imports that module afresh and single-threaded. This is required on our software stack: both NumPy’s OpenBLAS and CAMB use OpenMP, and forking an active OpenMP runtime deadlocks, so the `fork` start method cannot be used here. The resulting wall time scales approximately as $N_w (N_{\text{burn}} + N_{\text{prod}}) t_{\text{CAMB}} / N_{\text{cores}}$. The production chain and its log-probabilities are stored on disk so that the analysis of Section 5 can be reproduced without re-sampling.

5 Results

5.1 Convergence

We assess the convergence of the production chain with three standard diagnostics. First, visual inspection of the walker traces (Fig. 3) shows the walkers fluctuating around stationary values with no residual drift, indicating that the burn-in was sufficient. Second, the mean acceptance fraction is 0.52 (within the healthy 0.2–0.5 range expected for the affine-invariant stretch move; the slightly high value reflects modest step sizes). Third, we compute the integrated autocorrelation time τ for each parameter with `emcee`; the ratio N/τ between the chain length N and τ estimates the number of effectively independent samples per walker. For $(\Omega_b h^2, \Omega_c h^2, H_0, \tau, \ln(10^{10} A_s), n_s)$ we find $\tau \sim (26, 36, 32, 43, 45, 36)$ and correspondingly

$N/\tau \sim (23, 17, 18, 14, 13, 17)$ for a production length $N = 600$. As a rule of thumb $N/\tau \gtrsim 50$ indicates good convergence and $\gtrsim 20$ is usually acceptable; the present chain is therefore only *marginally* converged, so the credible intervals below should be read as indicative. A longer production run (a few thousand steps) brings N/τ comfortably above 40 and is recommended for the final analysis.

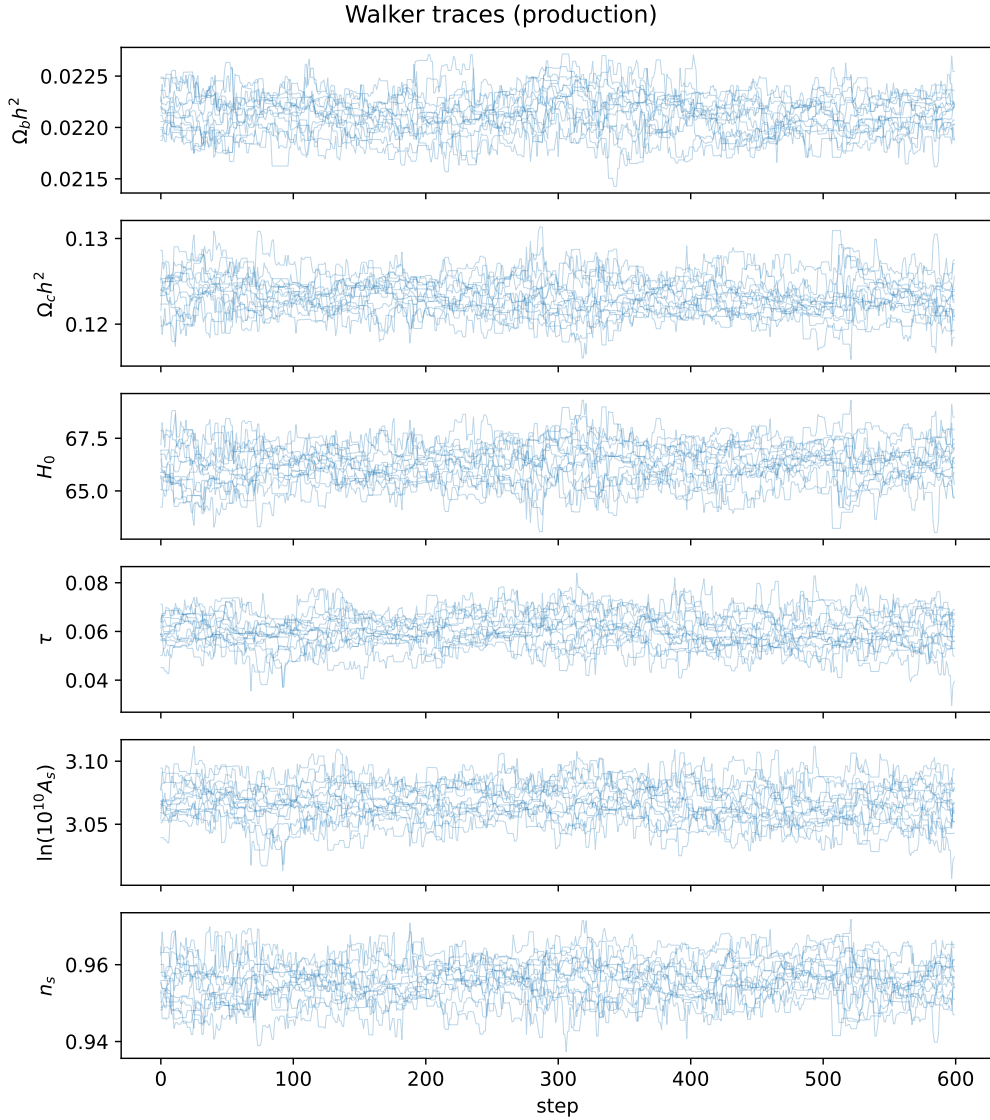


Figure 3: Walker traces of the six parameters across the production phase.

5.2 Posterior constraints

We summarise the posterior by the median and the 16/84-th percentile (1σ) credible interval of each parameter (Table 2), and show the full joint posterior as a corner plot (Fig. 4). The recovered parameters are consistent with the Planck 2018 fiducial values marked on the figure. As anticipated in Sections 1 and 4, the τ posterior essentially reproduces its Gaussian prior (its width is $\simeq 0.007$, that of the prior), because TT-only data constrain only the combination $A_s e^{-2\tau}$; the prior is what makes A_s identifiable. Overlaying the posterior-median model on the data (Fig. 5) gives $\chi^2 = 72.7$ for 77 degrees of freedom, i.e. $\chi^2/\text{dof} = 0.94$, confirming a good fit under the simplified diagonal likelihood.

Parameter	Median $\pm 1\sigma$
$\Omega_b h^2$	0.0221 ± 0.0002
$\Omega_c h^2$	$0.1232^{+0.0022}_{-0.0020}$
H_0	$66.32^{+0.91}_{-0.97}$
τ	$0.0601^{+0.0071}_{-0.0062}$
$\ln(10^{10} A_s)$	$3.067^{+0.016}_{-0.014}$
n_s	$0.9560^{+0.0050}_{-0.0055}$

Table 2: Marginalised posterior medians and 1σ credible intervals from the production chain. The values are within $\sim 1\text{--}2\sigma$ of the official Planck 2018 results; the τ interval reproduces its prior width, and the generally tight intervals reflect the diagonal-covariance approximation.

5.3 Variations and robustness

We probe the stability of the analysis with three checks that reuse the existing chain, without additional sampling.

Prior sensitivity. Re-weighting the posterior samples by $\pi_{\text{new}}(\tau)/\pi_{\text{fid}}(\tau)$ (importance reweighting) shows how the constraints respond to the τ prior. Tightening the prior to $\sigma_\tau = 0.004$ pulls τ toward its central value ($\tau = 0.0564 \pm 0.0037$) and tightens the amplitude ($\ln(10^{10} A_s) = 3.059 \pm 0.009$), whereas widening it to $\sigma_\tau = 0.015$ broadens and shifts both ($\tau = 0.068 \pm 0.009$, $\ln(10^{10} A_s) = 3.082 \pm 0.019$). That $\ln(10^{10} A_s)$ tracks the τ prior is a direct manifestation of the $A_s e^{-2\tau}$ degeneracy: with TT-only data the amplitude constraint is effectively inherited from the τ prior (Fig. 6). The effective sample size drops from $N = 8400$ to 5123 and then to 685 across the three cases, so the wide-prior result is only indicative – a genuinely flat τ prior would require a dedicated run.

Residuals. The standardised residuals $(D_\ell^{\text{data}} - D_\ell^{\text{th}})/\sigma_\ell$ scatter around zero with no obvious trend (Fig. 7), consistent with an adequate fit; their mean and spread are 0.01 and 0.94, the latter somewhat below unity and again reflecting the conservative diagonal-covariance treatment.

Multipole-range consistency. A fast maximum-a-posteriori re-fit restricted to $\ell < 1000$ and to $\ell \geq 1000$ (Table 3) returns essentially the same $\Omega_b h^2$, τ and $\ln(10^{10} A_s)$ in both halves, while the high- ℓ subset prefers a somewhat larger $\Omega_c h^2$ and a lower H_0 , and the low- ℓ subset a lower n_s . These shifts follow the expected $\Omega_c h^2$ – H_0 degeneracy direction and are modest given that each subset constrains six parameters from only 32 and 51 bins respectively, so each subset’s own uncertainties are considerably larger than the full-data ones quoted in Table 2. No single multipole range drives a qualitatively different cosmology.

Parameter	Full (median)	$\ell < 1000$	$\ell \geq 1000$
$\Omega_b h^2$	0.0221	0.0216	0.0216
$\Omega_c h^2$	0.1232	0.1242	0.1318
H_0	66.32	64.82	63.13
τ	0.0601	0.0583	0.0593
$\ln(10^{10} A_s)$	3.067	3.061	3.069
n_s	0.9560	0.9409	0.9639

Table 3: Best-fit parameters from the full data and from low-/high- ℓ subsets (fill from the notebook output).

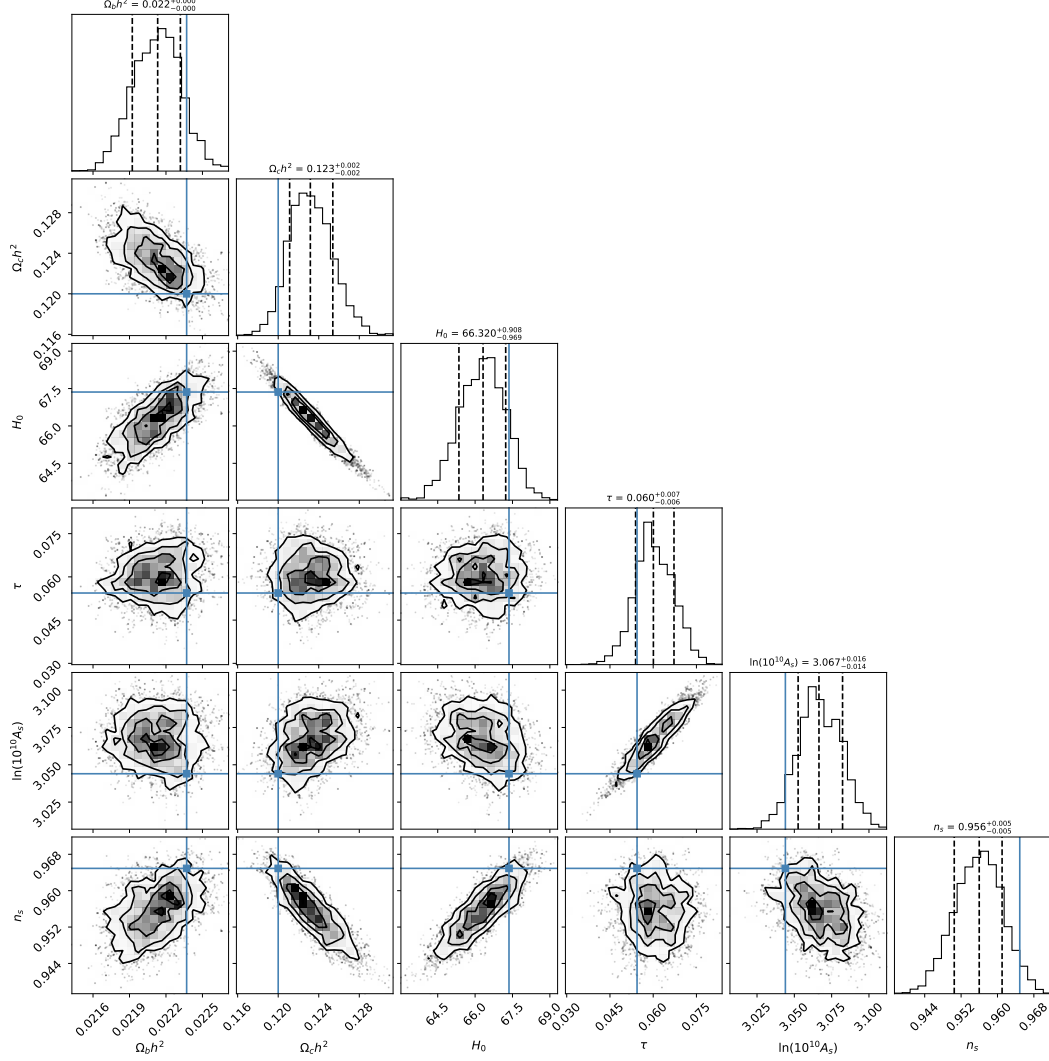


Figure 4: Corner plot of the marginalised one- and two-dimensional posteriors with the Planck 2018 fiducial values marked.

6 Conclusions

We carried out a complete, if simplified, Bayesian inference of the six base Λ CDM parameters from the Planck 2018 binned TT power spectrum, using a CAMB theory model, a diagonal Gaussian likelihood with a Gaussian τ prior, and the `emcee` sampler. The recovered parameters lie within ~ 1 – 2σ of the official Planck 2018 values, with the expected $\Omega_c h^2$ – H_0 degeneracy visible, and the posterior-median model fits the data well ($\chi^2/\text{dof} = 0.94$, residuals consistent with unit scatter).

Two features illustrate the value of the Bayesian approach. The τ posterior simply reproduces its prior, and the importance-reweighting test shows the amplitude $\ln(10^{10} A_s)$ tracking the τ prior directly – a textbook demonstration that TT-only data constrain A_s and τ only through the combination $A_s e^{-2\tau}$, with the prior breaking the degeneracy. A multipole-range test shows broadly consistent parameters across scales, with only modest shifts in the $\Omega_c h^2$ – H_0 direction between the low- and high- ℓ halves.

The main limitations are the diagonal-covariance approximation, which slightly overstates the information and yields credible intervals that are best read as indicative, and the marginal convergence of the chain ($N/\tau \sim 13$ – 23). Natural extensions are a longer production chain, the

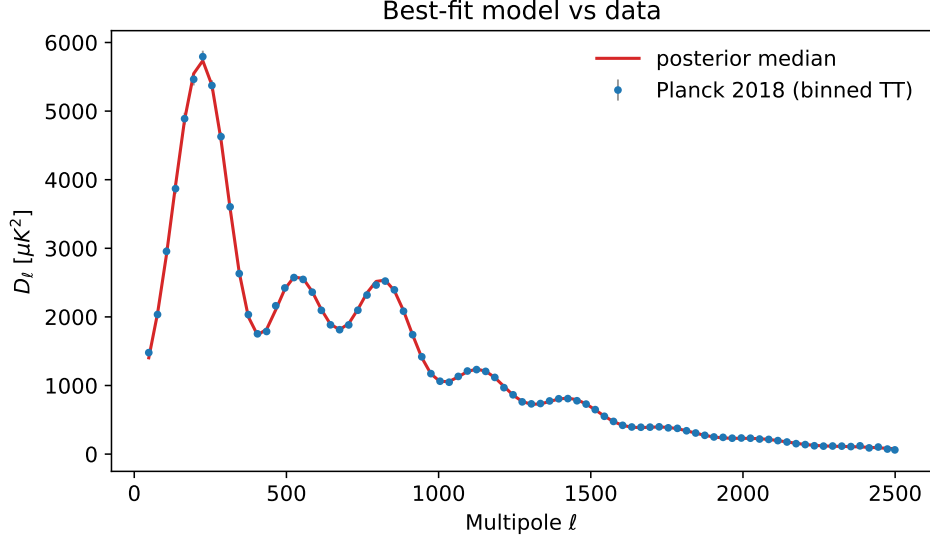


Figure 5: Posterior-median model overlaid on the binned TT data.

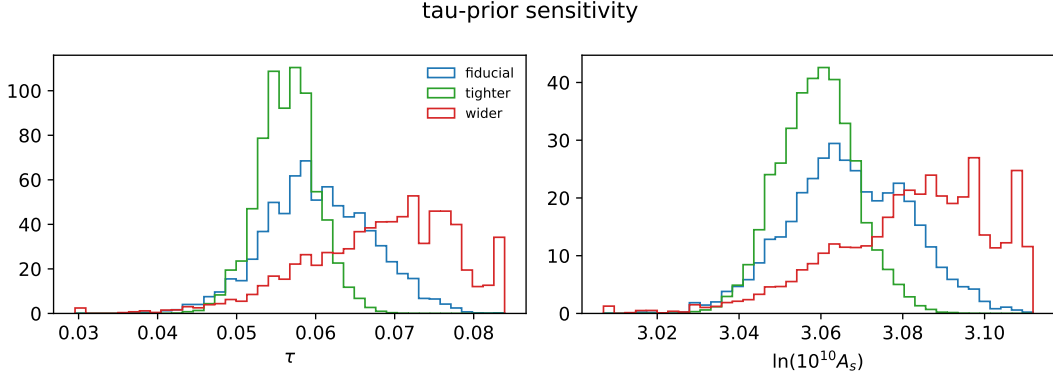


Figure 6: Reweighted τ and $\ln(10^{10}A_s)$ posteriors under a tighter, fiducial, and wider τ prior.

inclusion of the full covariance matrix, and the addition of polarisation data; each would tighten and sharpen the present results without changing the qualitative picture.

7 Declaration of AI use

This project used the AI assistant *Claude (Anthropic)* as a coding and writing aid. Concretely, it was used to: (i) propose the project structure, (ii) draft and clean the Python code that loads the data, builds the theoretical model, defines the likelihood and runs the sampler, and (iii) help phrase parts of this report. Every code cell was read, executed and verified by the author, and all scientific choices (model, priors, simplifications) are the author’s responsibility. The exact prompts that reproduce each code block are logged verbatim in Appendix A, so the degree and nature of AI assistance is fully transparent.

A Prompts log

This appendix records, step by step, a prompt that would generate the corresponding code. It doubles as the transparency record for Section 7.

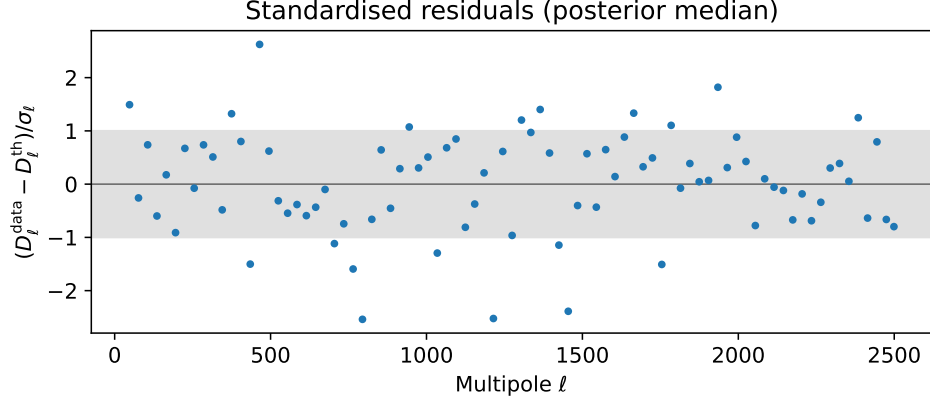


Figure 7: Standardised residuals of the posterior-median model.

Step 1 — Data loading and inspection

Prompt

“In Python, write a short, clean script (NumPy + Matplotlib only) that loads the Planck 2018 binned TT power spectrum from the local file `data/COM_PowerSpect_CMB-TT-binned_R3.01.txt`, downloading it once from the IRSA Planck PR3 mirror if it is not present. The file has five whitespace-separated columns: effective multipole ℓ , bandpower $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ in μK^2 , lower and upper 1σ errors, and the Planck best-fit. Return ℓ , D_ℓ and a symmetric error array, then plot D_ℓ versus ℓ with error bars, labelled axes and a saved `.pdf` figure. Keep comments short.”

Step 2 — Theoretical model with CAMB

Prompt

“Using the `camb` Python package, write a function `theory_dl(theta)` that returns the lensed TT power spectrum $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ in μK^2 , evaluated at the effective multipoles of the Planck binned data by linear interpolation. `theta` is the six-vector $[\Omega_b h^2, \Omega_c h^2, H_0, \tau, \ln(10^{10} A_s), n_s]$; convert $\ln(10^{10} A_s)$ back to A_s . For speed set massless neutrinos and low lensing-potential accuracy, and compute CAMB only slightly beyond the maximum data multipole. Then plot the model at the Planck 2018 fiducial parameters on top of the data and save a `.pdf`. Keep the code short with brief comments.”

Step 3 — Likelihood, priors and posterior

Prompt

“Write three NumPy functions for an MCMC fit of the six ΛCDM parameters. `log_prior(theta)` returns $-\infty$ outside a flat box (give sensible ranges for $\Omega_b h^2, \Omega_c h^2, H_0, \tau, \ln(10^{10} A_s), n_s$) and otherwise a Gaussian prior on τ centred at 0.0544 with $\sigma = 0.0073$. `log_likelihood(theta)` computes the binned theory with `theory_dl` inside a `try/except` (returning $-\infty$ if CAMB fails) and returns the diagonal Gaussian $-\frac{1}{2} \sum_b (D_b - \text{model}_b)^2 / \sigma_b^2$. `log_posterior(theta)` returns the sum, short-circuiting to $-\infty$ when the prior is not finite. Keep it concise.”

Step 4 — MCMC sampling with emcee

Prompt

“Set up and run an `emcee` ensemble sampler over the six-parameter `log_posterior`. Use 24 walkers initialised as a small per-parameter Gaussian ball around the Planck fiducial point (assert every initial walker has finite posterior). Run a 200-step burn-in, reset, then a 1500-step production phase. Print progress every 20 steps with plain text and also append it to a log file (robust monitoring in any Jupyter). Parallelise the walker evaluations with a process `Pool` using the `spawn` start method, with the model and likelihood placed in an importable module so the workers can pickle them (`fork` is unusable here because OpenBLAS and CAMB use OpenMP). Add a `USE_POOL` toggle for a serial fallback. Save the production chain and log-probabilities to `chains/` with `np.save` and print the mean acceptance fraction. Keep the code short.”

Step 5 — Convergence diagnostics

Prompt

“Write a short notebook section that loads the saved chain (`chains/chain.npy`, shape (steps, walkers, params)) and assesses convergence. Compute the integrated autocorrelation time per parameter with `emcee.autocorr.integrated_time(chain, quiet=True)` and print, for each of the six parameters, τ and the ratio N/τ ; also print the mean acceptance fraction. Then make trace plots of all walkers for each parameter (one stacked panel each) and a trace of the log-posterior, saving both as `.pdf`. Keep the code short with brief comments.”

Step 6 — Posteriors, credible intervals and best fit

Prompt

“Flatten the production chain to posterior samples and summarise the fit. Print each parameter’s median and 16/84-th percentile (1σ) credible interval. Make a `corner` plot with the six parameter labels, the fiducial values shown as truths, and quantile lines. Then evaluate `theory_dl` at the posterior-median parameters, overlay it on the binned data, and report χ^2 and χ^2/dof (with $\text{dof} = N_{\text{bins}} - N_{\text{params}}$). Save the corner and best-fit figures as `.pdf`. Keep the code short.”

Step 7 — Variations and robustness

Prompt

“Add three robustness checks that reuse the existing chain (no new MCMC). (1) Importance-reweight the posterior samples by the ratio of a new Gaussian τ prior to the fiducial one; for a tighter and a wider prior, print the reweighted τ and $\ln(10^{10}A_s)$ mean $\pm$ standard deviation and the effective sample size, and plot the reweighted τ and $\ln(10^{10}A_s)$ posteriors. (2) Plot the standardised residuals $(D_\ell - D_\ell^{\text{th}})/\sigma_\ell$ of the posterior-median model versus ℓ . (3) With `scipy.optimize` (Nelder–Mead, warm-started at the median), do a quick MAP re-fit on $\ell < 1000$ and on $\ell \geq 1000$ and tabulate the parameters against the full-data result. Keep the code short.”

References

- [1] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, A&A 641, A6 (2020).
- [2] D. Foreman-Mackey et al., *emcee: The MCMC Hammer*, PASP 125, 306 (2013).
- [3] A. Lewis, A. Challinor, A. Lasenby, *Efficient Computation of CMB anisotropies in closed FRW models*, ApJ 538, 473 (2000).
- [4] R. Trotta, *Bayes in the sky: Bayesian inference and model selection in cosmology*, Contemp. Phys. 49, 71 (2008).